3. Auto Arrange Details

One-line summary
Auto Arrange is a process that supplements unfilled arrangement slots through an internal selection when the number of required materials exceeds the available slots. The resulting entries are highly likely to affect final selection through Recursive Processing, Candidate Generation, and Performance-Source Contamination.

3-1. Occurrence Conditions

One-line summary
When the number of required materials, n, exceeds the three arrangement slots that can ordinarily be retained, arrangement information corresponding to the shortfall, $n - 3$, may be generated automatically.

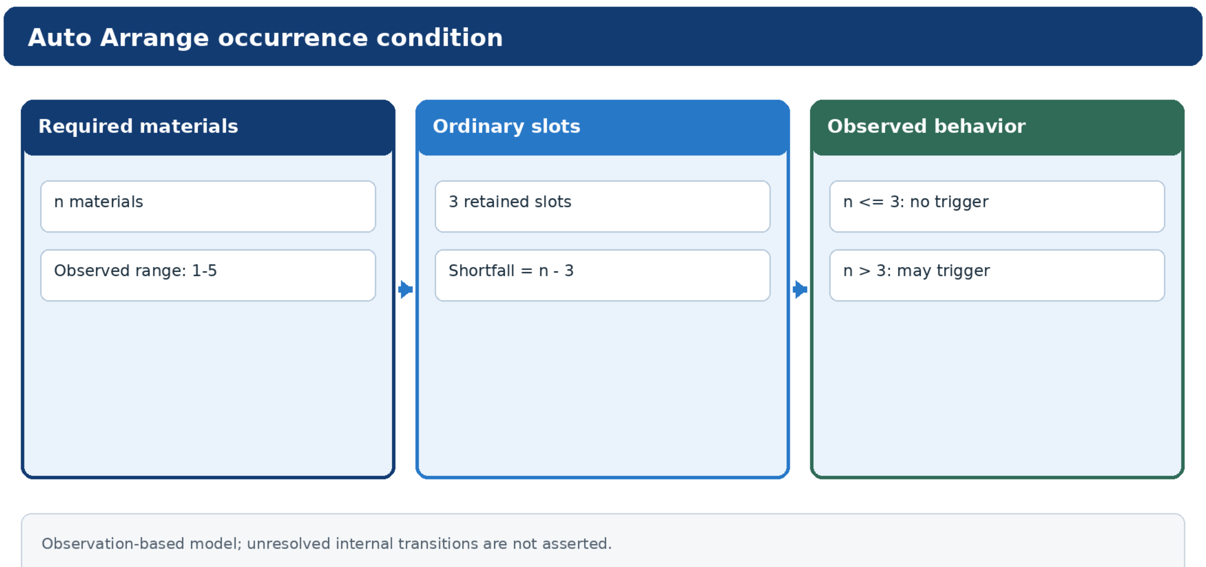


Figure 3-1. Auto Arrange occurrence condition.

Auto Arrange is an automatic arrangement-generation process that occurs when the number of materials required to craft the base equipment exceeds a certain value.

This document uses the following notation:

- number of required materials = n;
- number of arrangement slots that can ordinarily be retained = 3.

At least within the scope of the present observations, equipment for which:

$$n > 3$$

may automatically generate material entries corresponding to the shortfall:

$$n - 3.$$

The automatically generated arrangement information may not be merely temporary processing at the time of crafting; it may be retained inside the completed equipment.

It may therefore become involved in later processes, including:

- Recursive Processing;
- Candidate Generation;
- Performance-Source Contamination; and

03 Figure Identity / Numbering Record

Document identity patch; semantic authority is evaluated separately.

Figure 3-5	Random-selection observations and unresolved boundary
Figure 3-6	First, middle, and last positions observed
Figure 3-7	First-material inclusion
Figure 3-8	Full-pool selection consistency
Figure 3-9	RF4SP / RF5 observed differences
Figure 3-10	RF5 boundary comparison

Identity patch only: no new research claim is introduced.

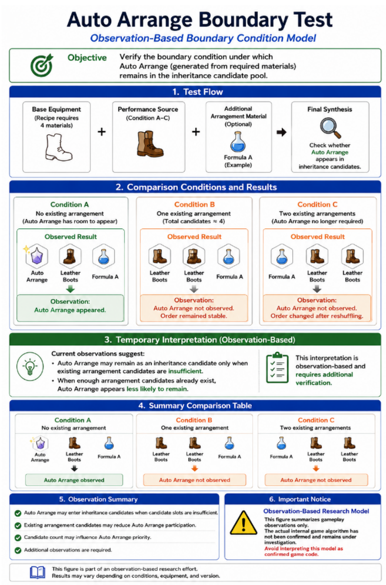
Figure 3-1

Placement 03-01 | Master M01-AA-OCCURRENCE | Japanese Canonical p.1

M01-AA-OCCURRENCE

01–05 CURRENT NORMALIZATION

Auto Arrange: Occurrence Is Observed, the Trigger Transition Is Not



OBSERVED CONDITION

Required-material count and empty-slot behavior can be recorded directly. RF5 behavior changes around tested three-entry / four-entry cases.

UNKNOWN TRANSITION

The game’s trigger test, pool construction, weighting, draw order, and implementation threshold are not directly observed.

Do not convert the observed checkpoint into an internal threshold.

Observed State → Unknown Transition → Observed State

01–05 Revision Master Set

Local caption

Auto Arrange occurrence condition

Current normalization: Observation, Model, and Unknown Transition remain explicitly separated.

- an increase in the Candidate Count.

Required Materials and Unfilled Slots

Number of required materials	Unfilled slots	Auto Arrange occurrence
1	0	No
2	0	No
3	0	No
4	1	Yes
5	2	Yes

3-2. Number of Required Materials and Number of Slots

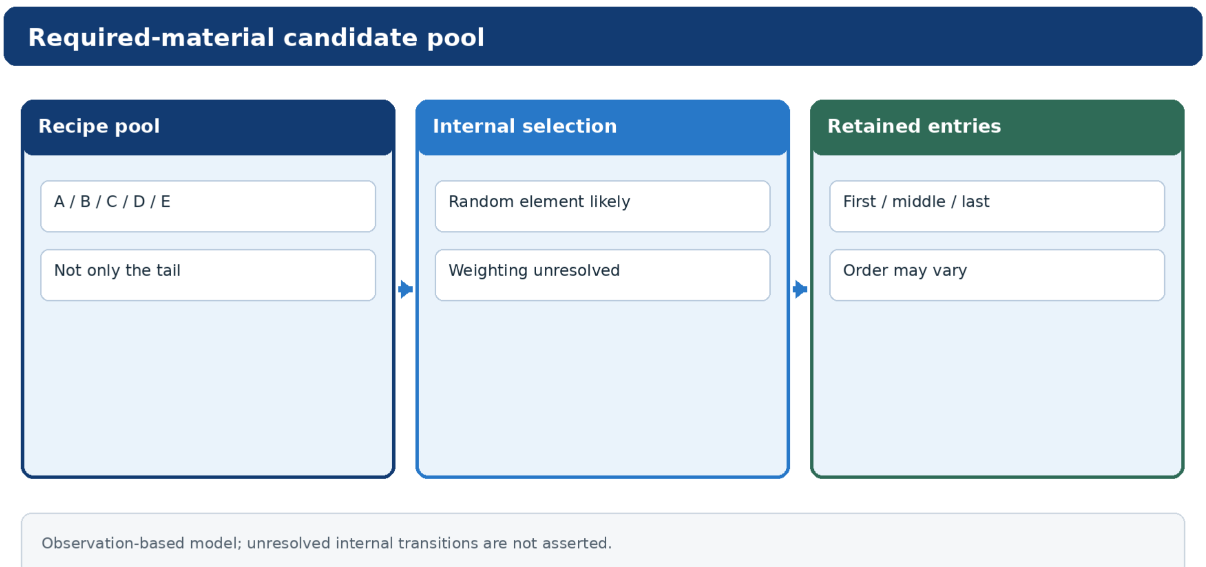


Figure 3-2. Required-material pool and unfilled slots.

One-line summary
Auto Arrange is treated as a process that supplements n - 3 entries, and its occurrence has been observed when n >= 4.

Figure 3-2

Placement 03-02 | Master M02-AA-SELECTION | Japanese Canonical p.2

M02-AA-SELECTION

01–05 CURRENT NORMALIZATION

Required-Material Pool: Inputs and Outputs Are Known, Selection Is Hidden

OBSERVED INPUT

Recipe and required materials
Visible empty slots
Additional arranged materials
Recorded source equipment

SUPPORTED BOUNDARY

Required-material pool behavior is supported by bounded observations. First, middle, and last recipe positions have appeared.

UNKNOWN TRANSITION

Selection unit
Weighting / priority
Draw order
Registration timing

OBSERVED OUTPUT

Completed equipment
Displayed inherited entries
Filled / unfilled slots
Recorded output order

NOT AUTHORIZED

“3 or fewer / 4 or more” may summarize observed RF5 cases; it is not proof of an internal branch.

NEXT CONTROL

Fix recipe and source; vary one visible candidate or position; retain every output.

Observed State → Unknown Transition → Observed State

01–05 Revision Master Set

Local caption

Required-material pool and unfilled slots

Current normalization: Observation, Model, and Unknown Transition remain explicitly separated.

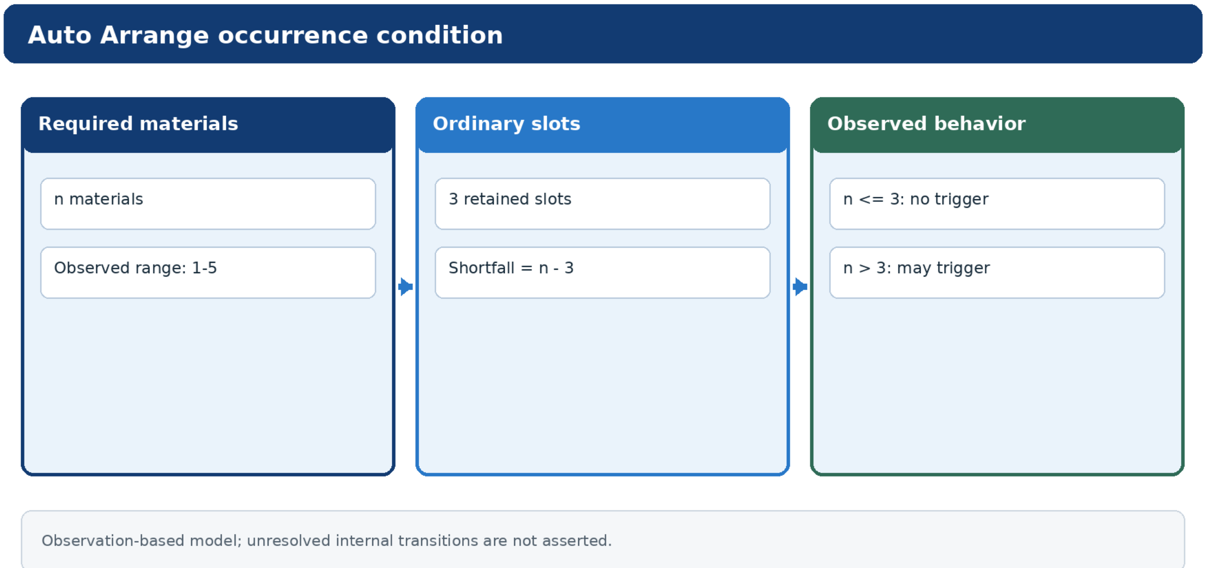


Figure 3-3. The n - 3 supplementation model.

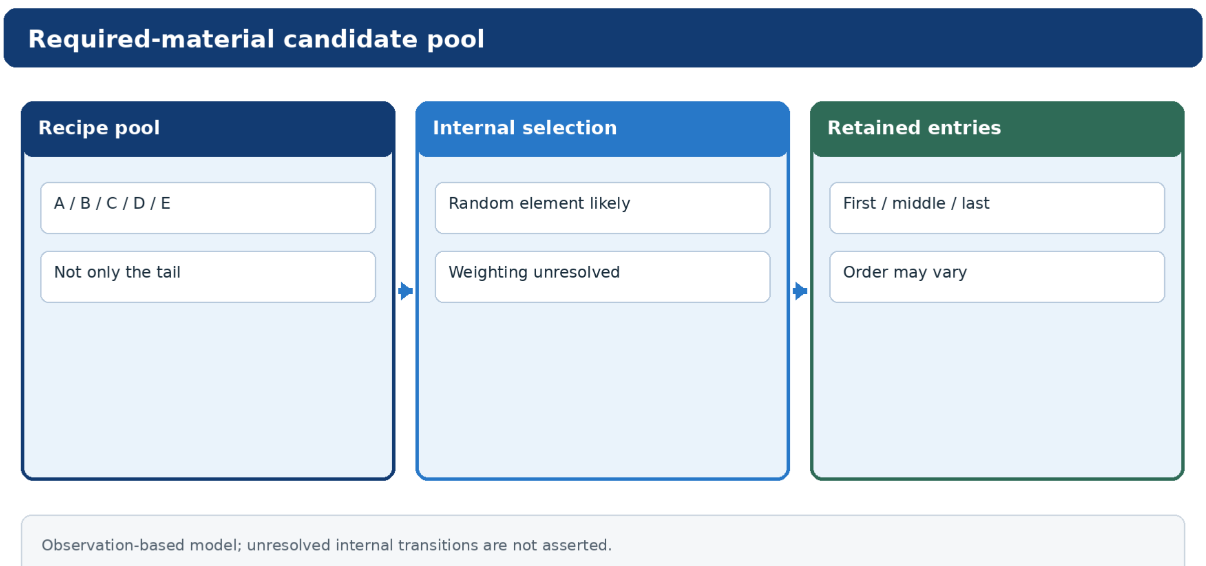


Figure 3-4. Selection is not limited to the end of the recipe list.

Under the current organization, Auto Arrange is treated as a process that supplements the difference between:

- the number of required materials; and
- the number of arrangement slots that can be retained.

For example:

Number of required materials, *n*	Number of unfilled slots, *n* - 3	Need for Auto Arrange
1	0	Unnecessary
2	0	Unnecessary
3	0	Unnecessary
4	1	May occur

Number of required materials, *n*	Number of unfilled slots, *n* - 3	Need for Auto Arrange
5	2	May occur

No equipment requiring six materials exists, so $n = 6$ is not considered. This applies to both RF4SP and RF5.

At least within the scope of the present observations, internal arrangement generation thought to originate from Auto Arrange has been observed in equipment requiring four or more materials.

Auto Arrange is unlikely to be a simple process that registers the excess materials from the end of the recipe-material list. It is highly likely to involve internal selection.

3-3. Random Selection Behavior

One-line summary
Automatically generated material entries are not fixed to the end of the list and tend to be selected from all required materials; changes in ordering or reselection may depend on the Candidate Count boundary.

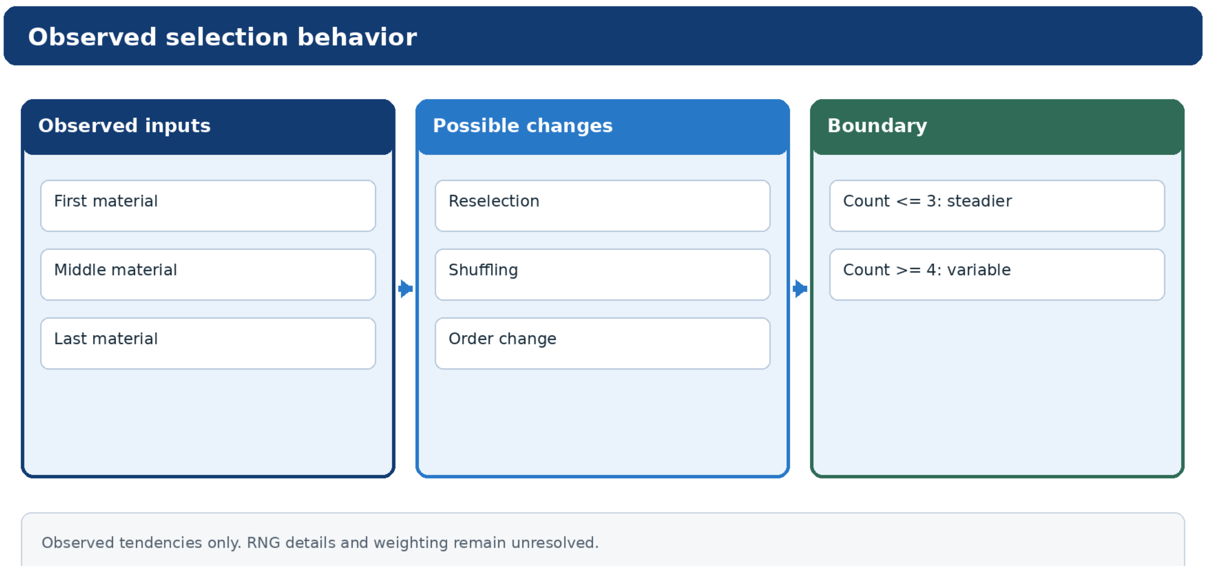


Figure 3-3. Random-selection observations and unresolved boundary.

At least within the scope of the present observations, Auto Arrange materials are highly likely to be selected with an element of randomness rather than in a fixed order.

In particular, the observations include:

- entries originating from the last material are not fixed;
- inclusion of a middle material; and
- inclusion of the first material.

These observations suggest that selection may occur from the full set of required materials.

In other words, when the required materials are:

A / B / C / D / E,

the process may select the required number of entries from all of A, B, C, D, and E, rather than always registering D and E.

In RF5, there may also be a behavioral difference between:

- a Candidate Count of three or fewer; and

- a Candidate Count of four or more.

With a Candidate Count of four or more, the following have been observed as possibilities:

- changes in ordering;
- reselection; and
- shuffling.

Observed Positions of Included Materials

Observation	Position of included material
Case 1	First
Case 2	Middle
Case 3	Last

However, the details of internal random-number processing (RNG) and any weighting process remain unresolved. No definitive claim is made at present.

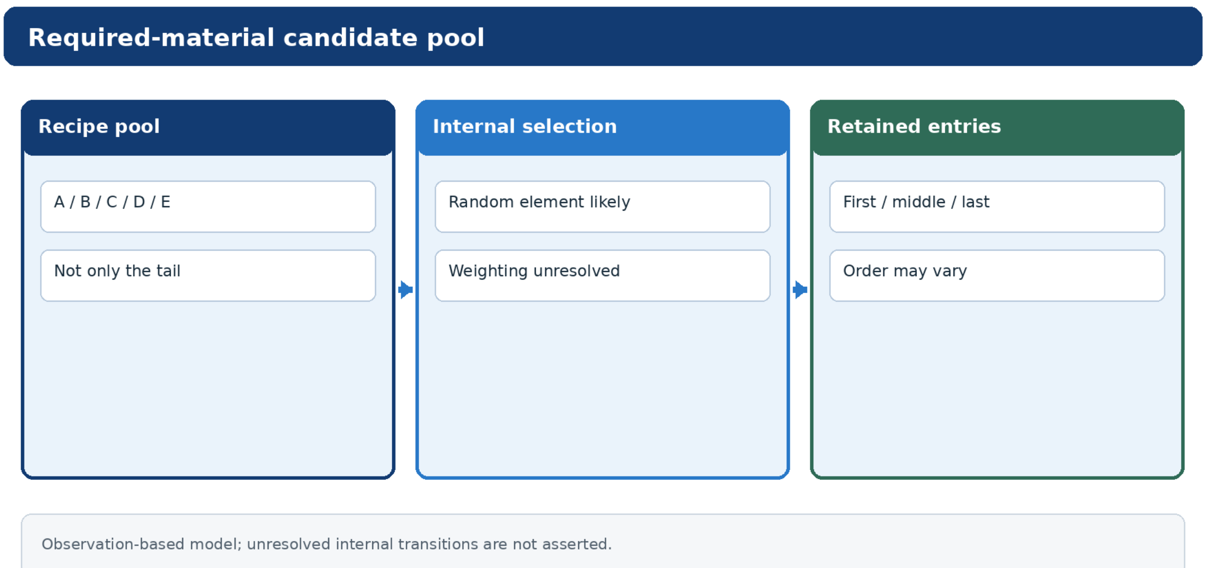


Figure 3-6. First, middle, and last positions have all been observed.

3-4. Inclusion of the First Material

One-line summary
The first and middle materials have been observed as Auto Arrange selections, making a simple “last-material rescue” model inconsistent with the observations.

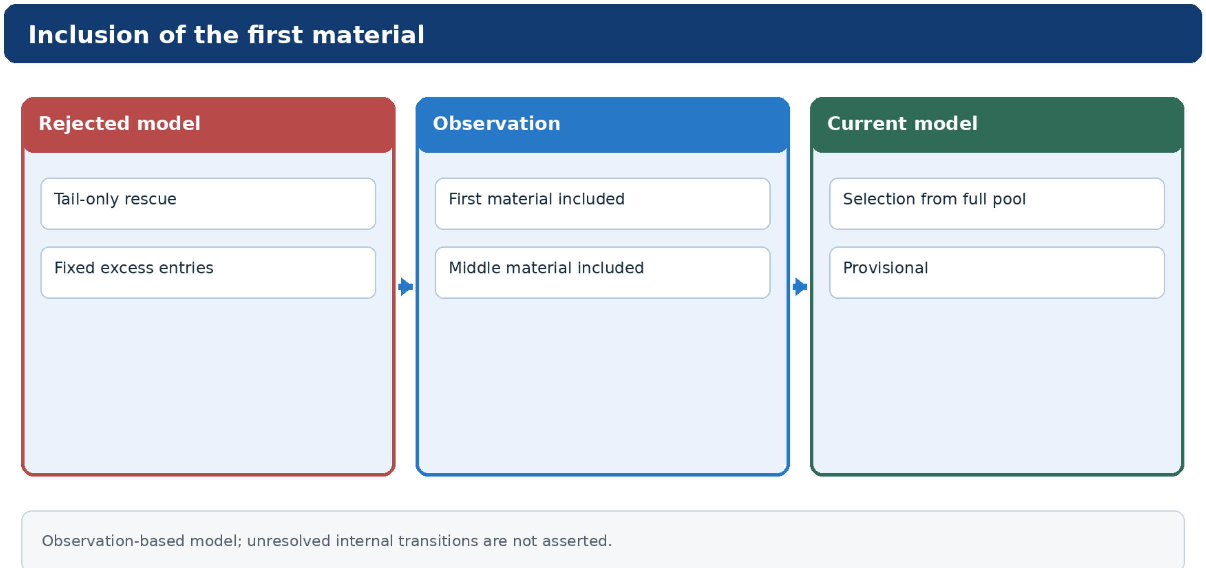


Figure 3-7. First-material inclusion contradicts a tail-only rescue model.

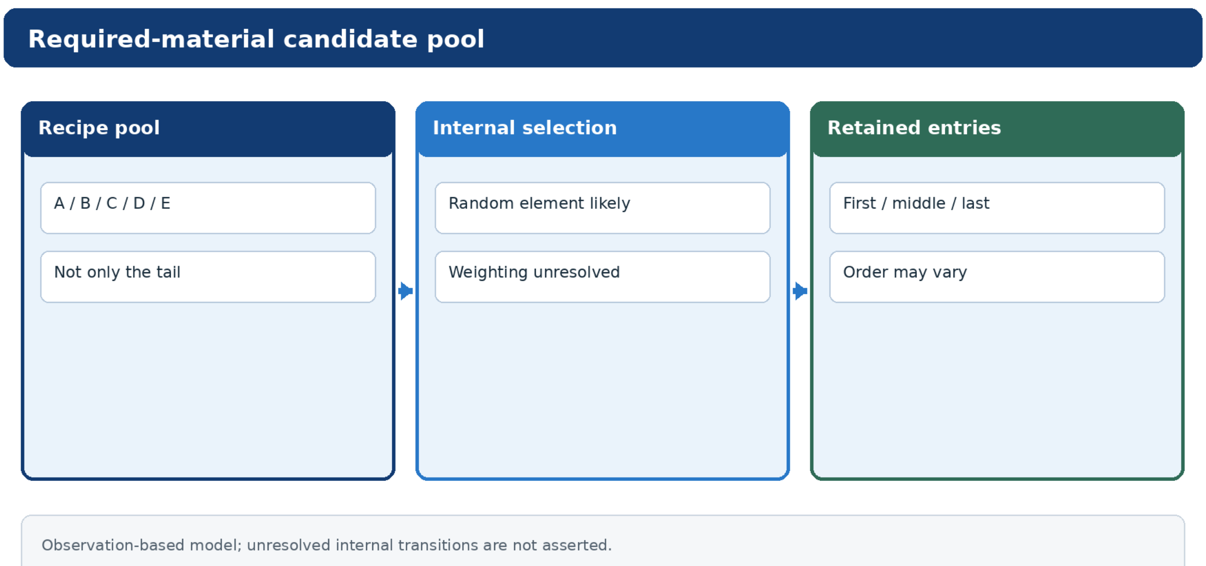


Figure 3-8. Full-pool selection is more consistent with the observations.

At an early stage, Auto Arrange was provisionally expected to register materials from the end of the required-material list automatically.

However, at least within the scope of the present observations, both of the following have now been confirmed:

- inclusion of the first material; and
- inclusion of a middle material.

Accordingly, interpreting Auto Arrange as a candidate-selection process covering all required materials is more consistent with the observed results than interpreting it as a process that merely rescues excess materials.

The inclusion of the first material is also important when considering its relationship with the following processes described later:

- Recursive Processing;
- Candidate Generation;

- Performance-Source Contamination; and
- an increase in the Candidate Count.

3-5. Differences Between RF4SP and RF5

One-line summary
Auto Arrange has been observed in both games, but RF5 may shift from relatively stable behavior toward shuffling at the Candidate Count boundary, while RF4SP may show stronger randomness from an earlier stage.

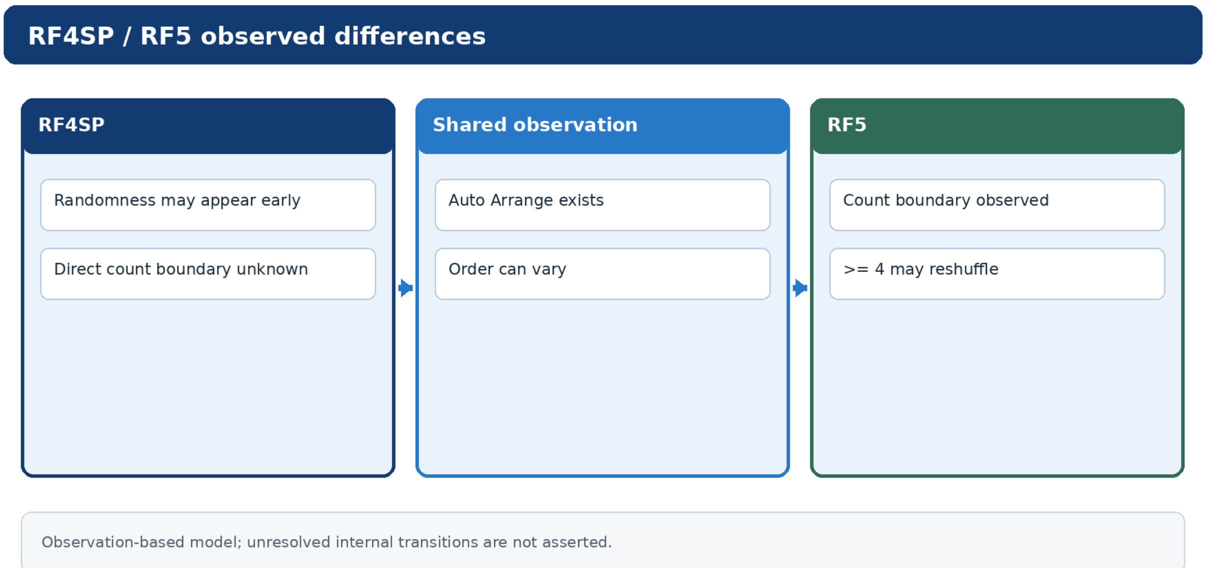


Figure 3-5. RF4SP / RF5 observed differences.

At present, Auto Arrange is highly likely to exist in both RF4SP and RF5. Differences may nevertheless exist in its internal behavior.

RF5

At least within the scope of the present observations, RF5 shows the following tendencies:

- behavior tending toward fixed ordering when the Candidate Count is three or fewer; and
- behavior tending toward shuffling or reselection when the Candidate Count is four or more.

The following have also been observed:

- changes in arrangement order;
- changes in candidate order; and
- retention of the source equipment's special effects.

These observations suggest that an increase in Candidate Count may strongly affect behavior.

Example Observation at the Candidate Count Boundary

Purpose

- Determine whether Auto Arrange remains in the selection candidate pool when the number of arrangement entries in the inheritance source increases.

Figure 3-8

Placement 03-08 | Master M03-CANDIDATE-BOUNDARY | Japanese Canonical p.11

M03-CANDIDATE-BOUNDARY

01–05 CURRENT NORMALIZATION

Candidate Count Organizes the Observation; It Does Not Reveal the Algorithm

VISIBLE SOURCES

Recipe materials
Source equipment history
Additional arranged materials
Auto Arrange state

REPLACEABLE MODEL

N = modeled candidate count
C(N,3) = modeled three-entry subsets
The model compares candidate-space growth.

UNKNOWN TRANSITION

Candidate construction
Selection unit
Weighting
Order handling

N = 3 MODEL STATE

C(3,3)=1 modeled subset. Stability in a bounded case does not establish an internal cutoff.

N = 4 MODEL STATE

C(4,3)=4 modeled subsets. Increased variation can be consistent with the model.

INTERPRETATION LIMIT

Full-pool and candidate-growth readings are better-fitting explanations, not confirmed mechanisms.

Observed State → Unknown Transition → Observed State

01–05 Revision Master Set

Local caption

Full-pool selection consistency

Current normalization: Observation, Model, and Unknown Transition remain explicitly separated.

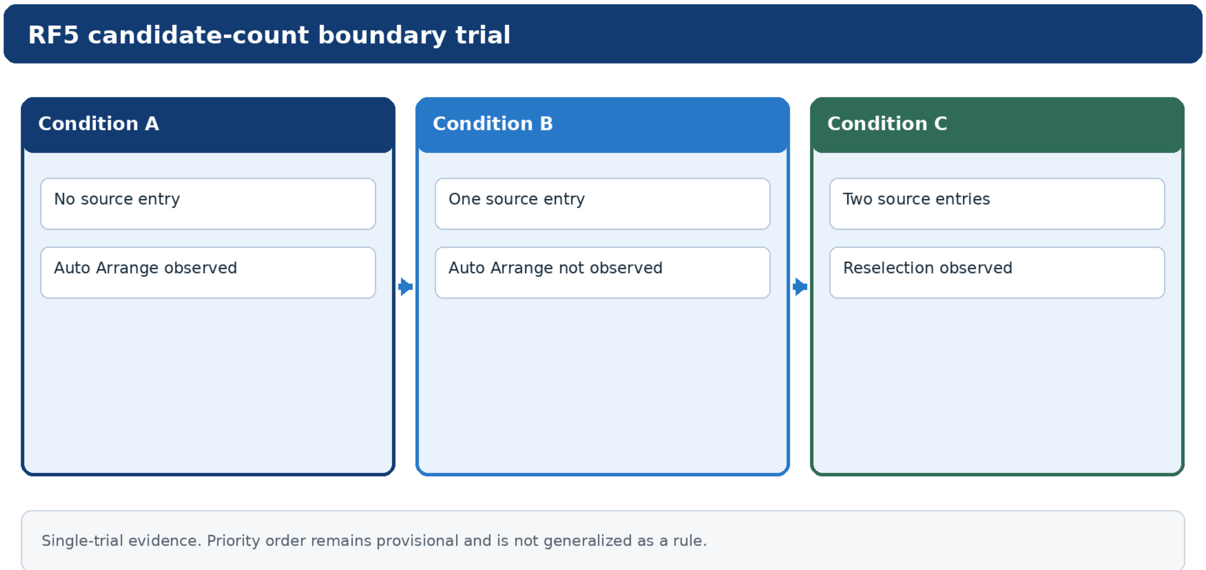


Figure 3-6. RF5 boundary comparison; single-trial evidence.

Conditions

- Conditions corresponding to a Candidate Count of four were used. Equipment examples include Step-In Boots and Strider Boots.
- The additional arrangement-slot material was Formuade.

Condition A: Inheritance source without an arrangement entry

Auto Arrange had room to enter the candidate pool.

Result

- Auto Arrange;
- Leather Boots; and
- Formuade.

Auto Arrange inclusion was confirmed.

Condition B: Inheritance source with one arrangement entry

The arrangement slots already corresponded to a count of four when Auto Arrange was included.

Result

- Leather Boots;
- Leather Boots; and
- Formuade.

Auto Arrange was not confirmed. No change in order occurred.

Condition C: Inheritance source with two arrangement entries

The arrangement entries alone already corresponded to a count of four, so Auto Arrange was unnecessary.

Result

- Formuade;
- Leather Boots; and
- Leather Boots.

Auto Arrange was not confirmed. The order changed, meaning that reselection itself did occur.

Provisional Interpretation

At least in this single trial, the results suggest that:

- Auto Arrange enters the candidate pool only when the inheritance source lacks sufficient arrangement entries; and
- when the inheritance source already provides enough arrangement candidates, Auto Arrange may have a low priority.

The number of trials is insufficient, so no definitive conclusion can be drawn.

Current Hypothesis - Provisional Because the Trial Count Is One

When the inheritance source has no existing arrangement entries:

- Auto Arrange may be more likely to enter the candidate pool.

When the inheritance source has existing arrangement entries:

- the source’s arrangement entries or its own performance-derived slot may be prioritized; and
- the Auto Arrange candidate may be displaced or assigned a lower priority.

In other words, a priority relationship may exist in the following form:

> Auto Arrange < existing arrangement entries in the inheritance source (or the source’s own performance-derived slot) = base arrangement entries

However, this remains non-definitive because the trial count is one. Additional testing is welcome.

RF4SP

Confirmation of Auto Arrange’s Existence

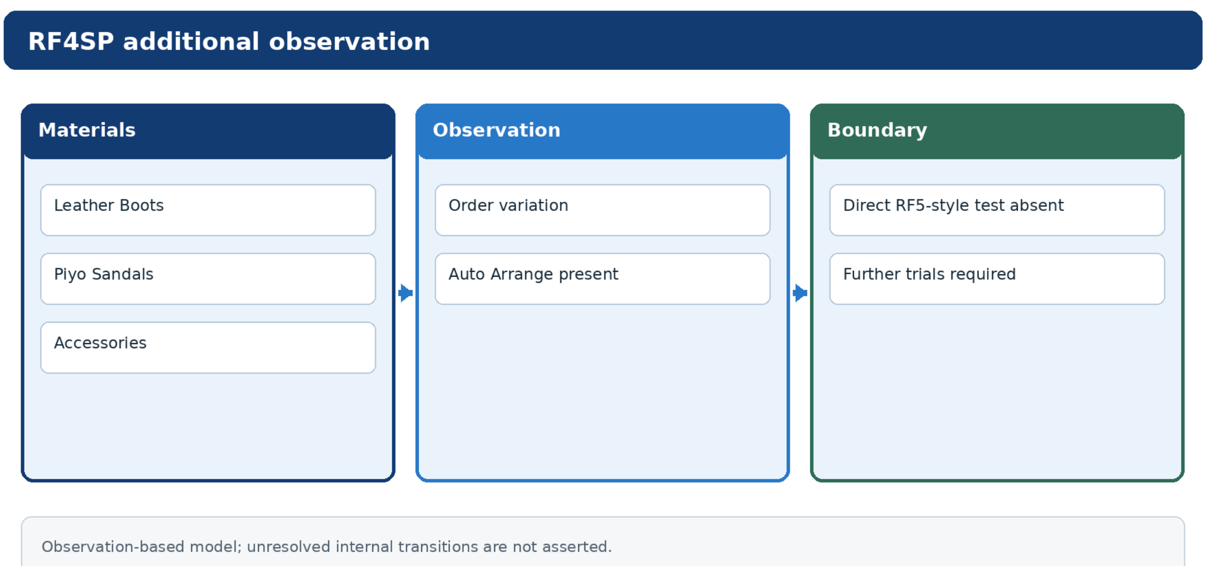


Figure 3-11. Additional RF4SP observation and its evidence boundary.

At least within the scope of the present observations, internal arrangement retention thought to originate from Auto Arrange has also been observed in RF4SP.

When crafting a Diamond Brooch, blue text indicating internally retained arrangement information was confirmed. Auto Arrange is therefore highly likely to exist in RF4SP.

However, the following detailed records are currently insufficient:

- which required material entered the arrangement information;
- the Candidate Count boundary; and
- changes in priority.

Tendency Toward Randomness

RF4SP may show stronger randomness than RF5 from the initial stage. The following have also been observed:

- Performance-Source Contamination; and
- dependence on the final arrangement slots.

Additional accessory-inheritance testing confirmed Performance-Source Contamination, which is explained in a later chapter, and changes in arrangement order.

The tests used:

- Cheap Bracelet;
- Bronze Bracelet;
- Silver Bracelet; and
- Gold Bracelet.

Inheritance was performed step by step using the equipment on the left as the inheritance source.

Examples of observed arrangement results were as follows.

After one inheritance stage

- Bronze Bracelet.

After two inheritance stages

- Cheap Bracelet;
- Bronze Bracelet.

After three inheritance stages

- Bronze Bracelet;
- Cheap Bracelet;
- Silver Bracelet.

After four inheritance stages

- After rechecking, this stage was judged unobserved.

At least within the scope of the present observations, not only Performance-Source Contamination but also the final arrangement order itself may be unfixed.

In particular, changes in ordering were observed between the two-stage and three-stage inheritance results under conditions thought to involve similar candidate groups. RF4SP may therefore exhibit stronger randomness than RF5 from an earlier stage.

However, the number of trials is insufficient. These results are treated as provisional observations at present.

Additional Testing Oriented Toward Recursive Processing

The additional test examined the treatment of inherited existing arrangement entries and internally retained arrangement information.

At least in this observation, the existing arrangement entries of the following items were retained:

- Leather Boots; and

- Feather Boots.

No additional inclusion of a required material originating from Feather Boots was confirmed.

Accordingly, when the following compete:

- inherited existing arrangement entries; and
- internally retained information originating from Auto Arrange,

the existing arrangement entries may be prioritized.

However, this RF4SP test differed from the Candidate Count boundary comparison conducted in RF5. It was oriented mainly toward observing:

- internally retained arrangement entries;
- recursive reference; and
- additional Candidate Generation.

Therefore, the RF5 observations of:

- behavioral change at a Candidate Count of four or more; and
- reduced priority of Auto Arrange,

were not directly confirmed in RF4SP.